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ЛАБОРАТОРНАЯ РАБОТА №4
“Приложения машинного обучения”

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1 Пайплайн Jenkins

Код для данной лабораторной работы был основан на коде сбора, обработки датасета и обучения модели из лабораторной работы №1.

Исходный код обучения модели, а также Jenkins-пайплайна доступен в репозитории.

Пайплайн состоит из 4 задач:

1. Download
2. Train model
3. Deploy
4. Healthcheck

1.1 Конфигурация пайплайна

```
pipeline {
  agent any

  stages {
    stage('Download') {
      steps {
        build job: 'Download'
      }
    }

    stage ('Train') {
      steps {
        build job: 'Train model'
      }
    }

    stage ('Deploy') {
      steps {
        build job: 'Deploy'
      }
    }

    stage ('Status') {
      steps {
        build job: 'Healthcheck'
      }
    }
  }
}
```

1.2 Работа пайплайна

S	W	Name ↓	Last Success	Last Failure	Last Duration	
✓	☁	ChocolateSales-Pipeline	57 min #7	59 min #6	1 min 16 sec	▶
✓	☀	Deploy	55 min #8	1 hr 34 min #1	58 ms	▶
✓	☀	Download	50 min #8	N/A	21 sec	▶
✓	☁	Healthcheck	55 min #7	58 min #5	77 ms	▶
✓	☀	Train model	56 min #8	1 hr 26 min #2	18 sec	▶

S

✓ < #7

Rerun

⚙

Started by admin Started 50 min ago Queued 12 ms Took 1 min 16 sec

Graph

Start Download Train Deploy Status End

Download 30s Train 26s Deploy 6s Status 10s

✓ Status 10s Started 49m ago Jenkins

Building Healthcheck 9s

0 Scheduling project: [Healthcheck](#)

1 Starting building: [Healthcheck #7](#)

2 Build [Healthcheck #7](#) completed: SUCCESS

1.3 Download

1.3.1 Управление исходным кодом

Git ?

Repositories ?

Repository URL ?

git@github.com:HowardLovekraft/UrFU-MLOps.git

Credentials ?

HowardLovekraft

+ Add

Advanced

1.3.2 Триггеры

Triggers

Set up automated actions that start your build based on specific events, like code changes or scheduled times.

- ☐ Trigger builds remotely (e.g., from scripts) ?
- ☐ Build after other projects are built ?
- ☐ Build periodically ?
- ☒ GitHub hook trigger for GITScm polling ?
- ☒ Poll SCM ?

Schedule ?

H/3 ****

Would last have run at Sunday, April 5, 2026, 7:34:00 PM Yekaterinburg Standard Time; would next run at Sunday, April 5, 2026, 7:37:00 PM Yekaterinburg Standard Time.

☐ Ignore post-commit hooks ?

1.3.3 Сборка

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute shell ?

Command

See [the list of available environment variables](#)

```
pwd
python3 -m venv ./my_env #создать виртуальное окружение в папку
. ./my_env/bin/activate #активировать виртуальное окружение
cd ./Lab_4 #перейти в директорию ./UrFU-ML0ps/lab4
python3 -m ensurepip --upgrade
pip3 install setuptools
pip3 install -r requirements.txt #установить пакеты python
python3 download_data.py #запустить python script
```

1.4 Train model

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute shell ?

Command

See [the list of available environment variables](#)

```
cd /var/lib/jenkins/workspace/Download
echo "Start train model"
. ./my_env/bin/activate #активировать виртуальное окружение
cd ./Lab_4 #перейти в директорию ./UrFU-ML0ps/lab4
python3 train_model.py > best_model.txt #обучение модели запись лога в файл best_model
```

1.5 Deploy

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute shell ?

Command

See [the list of available environment variables](#)

```
cd /var/lib/jenkins/workspace/Download
./my_env/bin/activate
cd ./Lab_4
export BUILD_ID=dontKillMe
export JENKINS_NODE_COOKIE=dontKillMe
path_model=$(cat best_model.txt)
mlflow models serve -m $path_model -p 5003 --no-conda &
```

1.6 Healthcheck

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute shell ?

Command

See [the list of available environment variables](#)

```
curl http://127.0.0.1:5003/invocations -H "Content-Type: application/json" -d '{"inputs": [[0.98544899, 0
```

```
Started by upstream project "ChocolateSales-Pipeline" build number 20
originally caused by:
  Started by user admin
Running as SYSTEM
Building in workspace /var/lib/jenkins/workspace/Healthcheck
[Healthcheck] $ /bin/sh -xe /tmp/jenkins1979194436420636767.sh
+ curl http://127.0.0.1:5003/invocations -H Content-Type: application/json -d '{"inputs": [[0.98544899, 0.32362852,
0.97384268, 0.29808986, 0.03494213, 0]]}'
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
  0     0     0      0     0      0      0      0  --:--:-- --:--:-- --:--:--     0
100   99  100    22  100    77    36   126  --:--:-- --:--:-- --:--:--   163
{"predictions": [0.0]}Finished: SUCCESS
```